Weighted cost function reconstruction in optical diffusion imaging

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ABSTRACT

A weighted distorted Born iterative method is presented for reconstructing optical diffusion images from scattering data. A generalization of the distorted Born iterative method that uses a preconditioned cost function and an elliptical constraint allows a weighting matrix to be applied to the gradient term in the iterative algorithm according to the reconstruction history. The proposed algorithm shows stable and fast convergence for reconstruction of high contrast inhomogeneities.

Keywords: optical diffusion imaging, distorted Born iterative method, weighting matrix, elliptical constraint, preconditioning, trust region, Levenberg-Marquardt method, image reconstruction

1. INTRODUCTION

The potential for optical imaging in highly scattering media such as tissue, as an alternative to X-ray or ultrasonic tomography, has been demonstrated.[1–5] In this technique, red or near infra-red light from a laser or light emitting diode is used. This low energy optical radiation presents significantly lower health risks than X-ray radiation. Also, suitable sources and detectors are relatively inexpensive, making such an instrument considerably cheaper than current computer tomography (CT) and magnetic resonance imaging (MRI) systems.

The nonlinearity and ill-posedeness of the inverse problem for optical imaging makes it more complicated to reconstruct images from scattering data than for the conventional imaging modalities. Hence, an iterative inversion approach is employed as a solution technique, since it can be applied to problems of any dimensionality and regularization[6] can easily be incorporated to help ameliorate the ill-posed nature of the problems by use of the trust region approach.[7,8] Consider a common algorithm form, as follows. Let $\mathbf{\Phi} = N(\mathbf{x})$ describe the nonlinear relationship between measurement at the detectors, $\mathbf{\Phi}$, and the unknown parameters, $\mathbf{x}$. First, the Jacobian matrix $J = \partial\Phi/\partial\mathbf{x}$ is derived at each iteration, which describes the variation of the measurement with respect to a small variation of the parameters. A perturbation in $\mathbf{\Phi}$, $\mathbf{\Delta\Phi}$, is related to a change in $\mathbf{x}$, $\mathbf{\Delta x}$, by $\mathbf{\Delta\Phi} \simeq J \cdot \Delta\mathbf{x}$. Second, a trust region where the perturbation model holds is defined as a constraint. In the Levenberg-Marquardt method, the trust region is defined as $||\mathbf{\Delta x}|| \leq \rho^n$, where ρ^n is the radius at the n-th step. This formulation is in fact equivalent to the distorted

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Born iterative method (DBIM).[9,10] If a regularization parameter λ is updated appropriately, the Levenberg-Marquardt method or DBIM is known to have guaranteed convergence to a local minimum. However, the regularization step is quite critical for satisfactory solution of inverse problems. An optimal value of λ can be obtained from another optimization problem, which significantly increases the computational burden.

We propose a weighting technique as an alternative to the conventional regularization method. In this technique, that we call the weighted distorted Born iterative method (WDBIM), a diagonal weighting matrix is used as a regularization term, and is automatically adapted from the reconstruction history. Since the diagonal weighting matrix is determined automatically, the additional computational burden for regularization is negligible. In this paper, we explore this approach and show simulated reconstructions for several inhomogeneity geometries, including a tissue phantom model.

2. DISCRETE NORMALIZED INTEGRAL EQUATION

Consider a 2-D experimental domain Ω with a unit line source located at $\vec{p}$ and the resulting fluence measured at $\vec{q}$, as shown in Fig. 1. The line source is oriented along the z-axis, and $\vec{r} = x\hat{x} + y\hat{y}$ defines a point in the plane. In the following, the total number of sources is denoted by N_s and the number of receivers (for which data are collected by each source location) N_r. Let $\mu_a(\vec{r})$ and $\mu_s'(\vec{r})$ represent the

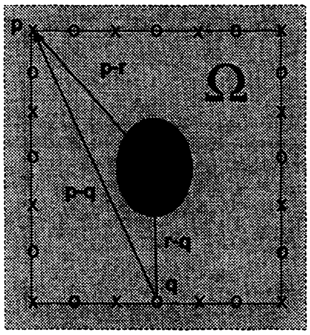

Figure 1. Data acquisition geometry with sources and detectors uniformly located around the boundary. The modulation frequency is 200MHz. The background properties are $\mu_a^b = 0.02cm^{-1}$ and $\mu_s'^b = 10.0cm^{-1}$

unknown absorption and equivalent-isotropic scattering coefficients in the medium (with units of cm^{-1}), and $D(\vec{r}) = [3(\mu_a(\vec{r}) + \mu_s'(\vec{r}))]^{-1}$ be the diffusion coefficient. The frequency domain diffusion equation for the fluence $\phi(\vec{r})$ (Wcm^{-2}), due to a sinusoidally intensity modulated source of light ($\sim \exp(-i\omega t)$), is given by[5]

$$\begin{aligned}\nabla^2\phi(\vec{r}) + k^2(\vec{r})\phi(\vec{r}) &= -s(\vec{r}) - \frac{\nabla D(\vec{r})\cdot\nabla\phi(\vec{r})}{D(\vec{r})}\\ k^2(\vec{r}) &= -\frac{\mu_a(\vec{r})}{D(\vec{r})} + \frac{i\omega}{cD(\vec{r})},\end{aligned} \tag{1}$$

where $k^2(\vec{r})$ is the wavenumber squared, c is the speed of light in the medium and $s(\vec{r})$ is the source signal. We neglect $\nabla D(\vec{r})\cdot\nabla\phi(\vec{r})$ for simplicity, which introduces a small error.[5]

Consider the actual medium parameters as a perturbation of a background medium, with absorption and scattering coefficients described by $\mu_a^b(\vec{r})$ and $\mu_s^{b'}(\vec{r})$, respectively, and an associated diffusion coefficient $D^b(\vec{r}) = [3(\mu_a^b(\vec{r}) + \mu_s^{b'}(\vec{r}))]^{-1}$ and squared wavenumber $k_b^2(\vec{r}) = -\mu_a^b(\vec{r})/D^b(\vec{r}) + i\omega/cD^b(\vec{r})$. Let $f(\vec{r}) = k^2(\vec{r}) - k_b^2(\vec{r})$ be the contrast between the two squared wave numbers. Then, the total fluence is represented as a scalar integral equation of the second kind:[5]

$$\begin{aligned} \phi(k, \vec{p}_l, \vec{q}_m) &= \phi_i(k_b, \vec{p}_l, \vec{q}_m) + \int_\Omega d\Omega_r G(k_b, \vec{r}, \vec{q}_m)\phi(k, \vec{p}_l, \vec{r}) f(\vec{r}) \\ l &= 1 \cdots N_s; \qquad m = 1 \cdots N_r, \end{aligned} \tag{2}$$

where $G(k_b, \vec{r}, \vec{q}_m)$ is the Green's function with background parameters $k_b(\vec{r})$, ϕ_i is the incident fluence (the total fluence if the medium were identical to the background), ϕ is the total fluence, $\vec{p}_l$ denotes the source location and $\vec{q}_m$ is the detector location. With an updated $k_b(\vec{r})$, the Green's function for an inhomogeneous medium will be required and hence must be calculated numerically. Note that since the unknown value ϕ depends on $f(\vec{r})$, (2) is a nonlinear equation. The goal is to find a $f(\vec{r}) = k^2(\vec{r}) - k_b^2(\vec{r})$ that satisfies (2). After obtaining $f(\vec{r})$, the change in absorption and scattering coefficient from the background values can be recovered.[5]

The magnitude of the incident fluence is very dependent on the source-detector separation, being larger for the detectors closer to the source. As shown in Fig. 1, the detectors are considered here to be located uniformly along the boundary, hence the magnitude of the incident fluence at locations near the source will be larger than those at locations further away. We therefore consider the following normalized integral equation for the scattered fluence:

$$\begin{aligned} \psi_s(k, \vec{p}_l, \vec{q}_m) &= \frac{\phi(k, \vec{p}_l, \vec{q}_m) - \phi_i(k_b, \vec{p}_l, \vec{q}_m)}{\phi_i(k_b, \vec{p}_l, \vec{q}_m)} \\ &= \frac{1}{\phi_i(k_b, \vec{p}_l, \vec{q}_m)} \int_\Omega d\Omega_r G(k_b, \vec{r}, \vec{q}_m) f(\vec{r}) \phi(k, \vec{p}_l, \vec{r}) \\ l &= 1 \cdots N_s; \qquad m = 1 \cdots N_r \quad . \end{aligned} \tag{3}$$

Using (3) makes the contribution of the incident fluence uniform irrespective of the source-detector distance, thereby reducing the ill-posedness resulting from the variation in distance.

The method of moments procedure with pulse-basis functions[9] can be applied to discretize (3) into N cells, with point matching being a test function that leads to a matrix equation. The same pulse-basis functions are used for the contrast $f(\vec{r})$ and the total fluence. We assume here that the contrast profile is constant over each cell, giving the discrete approximation to (3) as

$$\mathbf{\Phi_s} = P^o(\mathbf{x}) \cdot \mathbf{x} \tag{4}$$

with

$$\mathbf{\Phi_s} = \begin{bmatrix} \psi_s(k, \vec{p}_1, \vec{q}_1) \\ \psi_s(k, \vec{p}_1, \vec{q}_2) \\ \vdots \\ \psi_s(k, \vec{p}_{N_s}, \vec{q}_{N_r}) \end{bmatrix}$$

$$P^o(\mathbf{x}) = \begin{bmatrix} (\mathbf{p}_{1,1})^T \\ (\mathbf{p}_{1,2})^T \\ \vdots \\ (\mathbf{p}_{N_s,N_r})^T \end{bmatrix}$$

$$
\begin{aligned}
\mathbf{p}_{i,j} &= \begin{bmatrix} \frac{G(k_b;\vec{r}_1;\vec{q_j})\phi(k;\vec{p}_i;\vec{r}_1)}{\phi_i(k_b;\vec{p_i};\vec{q_j})}\delta\Omega_1 \\ \frac{G(k_b;\vec{r}_2;\vec{q_j})\phi(k;\vec{p}_i;\vec{r}_2)}{\phi_i(k_b;\vec{p_i};\vec{q_j})}\delta\Omega_2 \\ \vdots \\ \frac{G(k_b;\vec{r}_N;\vec{q_j})\phi(k;\vec{p}_i;\vec{r}_N)}{\phi_i(k_b;\vec{p_i};\vec{q_j})}\delta\Omega_N \end{bmatrix} \\
\mathbf{x} &= [f(\vec{r}_1)\cdots f(\vec{r}_N)]^T \quad ;
\end{aligned} \tag{5}
$$

i.e., $P^o(\mathbf{x})$ is an $(N_s \times N_r) \times N$ matrix of complex valued elements that are a function of the total fluence, ϕ; ϕ is a function of $\mathbf{x}$, and $\delta\Omega_m$ is the area of the m-th cell. Note that the superscript o denotes that the k^b is used to compute the Green's function. The N-dimensional vector $\mathbf{p}_{i,j}$ in (5) applies for source i and detector j.

3. WEIGHTED DISTORTED BORN ITERATIVE METHOD

To provide additional control of the iterative process, over and above a single regularization parameter, λ, we develop a preconditioned cost function approach that allows an improvement in the condition number of $P(\mathbf{x})$, thereby allowing an improvement in the convergence properties of the usual DBIM algorithm. The procedure involves two steps. The first is a preconditioning step, where the goal is to make the non-zero singular values have similar size, thereby reducing the condition number. The second step involves adding a regularization term to compensate for the zero singular values, as is done in nonlinear optimization techniques for ill-posed problems.

In the weighted distorted Born iterative method, the optimization problem is

$$
\text{minimize} \qquad \| \, M(\mathbf{x})P^o(\mathbf{x}) \cdot \mathbf{x} - M(\mathbf{x})\mathbf{\Phi_s} \, \|, \tag{6}
$$

where $M(\mathbf{x})$ is a preconditioning matrix,

$$
M(\mathbf{x}) \;=\; (P^o(\mathbf{x})P^o(\mathbf{x})^H + \lambda I)^{-\frac{1}{2}}. \tag{7}
$$

Note that a small positive λ is inserted to make the matrix invertible. The preconditioner, $M(\mathbf{x})$, makes the singular values of $M(\mathbf{x})P(\mathbf{x})$ nearly unity, thereby yielding fast convergence. However, direct application of the Newton method for (6) is nearly impossible, because of the computational burden of computing the derivatives of $M(\mathbf{x})$ with respect to $\mathbf{x}$. Hence, we use an approximate constant preconditioning matrix $\hat{M}(\mathbf{x}^n)$ at the n-th iteration.

The linearized optimization problem after the n-th iteration is

$$
\text{minimize} \qquad \| \, \hat{M}(\mathbf{x}^n)P(\mathbf{x}^n) \cdot \mathbf{\Delta x} - \hat{M}(\mathbf{x}^n)\mathbf{\Phi_s}^n \, \|, \tag{8}
$$

where $P(\mathbf{x}^n)$ is the Jacobian of (4) at $\mathbf{x} = \mathbf{x}^n$, and

$$
\begin{aligned}
\mathbf{\Phi}_s^n &= \mathbf{\Phi}_s - P^o(\mathbf{x}^n) \cdot \mathbf{x}^n \\
\hat{M}(\mathbf{x}^n) &= (P(\mathbf{x}^n)P(\mathbf{x}^n)^H + \lambda I)^{-\frac{1}{2}}.
\end{aligned} \tag{9}
$$

The Jacobian $P(\mathbf{x}^n)$ is given by:

$$
P(\mathbf{x}^n) \;=\; \begin{bmatrix} (\mathbf{p}_{1,1})^T \\ (\mathbf{p}_{1,2})^T \\ \vdots \\ (\mathbf{p}_{N_s,N_r})^T \end{bmatrix}
$$

$$\mathbf{p}_{i,j} = \begin{bmatrix} \frac{G(k^n, \vec{r}_1, \vec{q_j})\phi(k, \vec{p}_i, \vec{r}_1)}{\phi_i(k^n, \vec{p_i}, \vec{q_j})}\delta\Omega_1 \\ \frac{G(k^n, \vec{r}_2, \vec{q_j})\phi(k, \vec{p}_i, \vec{r}_2)}{\phi_i(k^n, \vec{p_i}, \vec{q_j})}\delta\Omega_2 \\ \vdots \\ \frac{G(k^n, \vec{r}_N, \vec{q_j})\phi(k, \vec{p}_i, \vec{r}_N)}{\phi_i(k^n, \vec{p_i}, \vec{q_j})}\delta\Omega_N \end{bmatrix} . \tag{10}$$

Note the difference between (10) and (5). In (10), the Green's function is updated with the previous reconstruction k^n. In (5), the Green's function is determined from the initial estimate of the background parameter, k_b.

In the conventional Levenberg-Marquardt method, the trust region approach[7,8] is used to guarantee the invertibility of the Hessian and the descent property[11] of (6). The trust region, where the quadratic model (8) holds, works as a circular constraint,

$$||\mathbf{\Delta x}|| \leq \rho^n, \tag{11}$$

where ρ^n is a radius of the trust region after the n-th step. However, in the WDBIM, we use the following elliptical constraint,

$$\mathbf{\Delta x}^H (W^n)^{-1} \mathbf{\Delta x} \leq r^n \;, \tag{12}$$

where r^n determines the size of the ellipse after n-th iteration. The constraint (12) corresponds to restricting the reconstruction, $\mathbf{\Delta x}$, inside an ellipse whose axial ratio is determined by W^n. We choose

$$W^n(i,j) = \left(\frac{|\mathbf{x}^n(i)|}{||\mathbf{x}^n||_\infty}\right)^q \delta(i,j), \tag{13}$$

where the superscript q, the order of the constraint, is a power and $\delta(i,j)$ is the Kronecker delta. Note that if $q = 0$ the reconstruction is restricted to the inside of a circle, which corresponds to the constraint of the DBIM. If $q > 0$ is used, the reconstruction is restricted to the inside of an ellipse, which has its longer axis in the direction of the largest change in the previous reconstruction. A constraint with $q > 0$ is preferable because the output response is usually spatially low-pass filtered,[9] and the region of support for a spatially low-pass filtered signal is larger than that of the original signal in the spatial domain. Using $q > 0$ reduces the spatial support of the reconstructed result. In this paper, we compare two different orders of constraint, $q = 0$ and $q = 1$.

The solution for (8) with constraint (12) is

$$\mathbf{x}^{n+1} = \mathbf{x}^n + G^{-1}(P^H(\mathbf{x}^n)P(\mathbf{x}^n) + \lambda I)^{-1}P^H(\mathbf{x}^n)\mathbf{\Phi_s}^n, \tag{14}$$

where G is the Hessian matrix given by

$$G = P^H(\mathbf{x}^n)(P(\mathbf{x}^n)P^H(\mathbf{x}^n) + \lambda I)^{-1}P(\mathbf{x}^n) + \beta^n(W^n)^{-1} \tag{15}$$

and β^n is a Lagrangian constant for the constrained optimization problem.[11] If β^n is large, the approximate Hessian matrix is,

$$G \simeq \beta^n(W^n)^{-1}. \tag{16}$$

Therefore, the quasi-Newton update is

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \frac{1}{\beta^n}W^n(P^H(\mathbf{x}^n)P(\mathbf{x}^n) + \lambda I)^{-1}P^H(\mathbf{x}^n)\mathbf{\Phi_s}^n. \tag{17}$$

If $W^n = I$ and $\beta^n = 1$, the update equation (17) is of the same form as that for the DBIM.

4. COMPUTATIONAL EFFICIENCY

The computational complexity has been reduced by the following techniques. First, we use the reciprocity principle[9] to reduce the number of forward solutions at each iteration. As a Jacobian of (4) at $\mathbf{x} = \mathbf{x}^n$, we need to compute the matrix $P(\mathbf{x}^n)$. For the i-th source and j-th receiver pair, the corresponding element of $P(\mathbf{x}^n)$, after the n-th estimate, is given by (10). Note that at each iteration, we need to update the Green's function from the previous $\mathbf{x}^n$. The time consuming part of the computation lies in computing the Green's function $G(k^n, \vec{r}_m, \vec{q}_j), m = 1 \cdots N, j = 1 \cdots N_r$. To compute G, we need to put a source at $\vec{r} = \vec{r}_m$ and compute the fluence at $\vec{r} = \vec{q}_j$. This requires N forward solutions to fill the matrix $P(\mathbf{x}^n)$. However, the reciprocity theorem[9] tells us that $G(k^n, \vec{r}_m, \vec{q}_j) = G(k^n, \vec{q}_j, \vec{r}_m), m = 1 \cdots N, j = 1 \cdots N_r$. Hence, we can put sources at $\vec{r} = \vec{q}_j$, thereby requiring only N_r forward solutions. Second, in computing (17), we can reduce the complexity by the following matrix relationship[12]:

$$(P^H(\mathbf{x}^n)P(\mathbf{x}^n) + \lambda I)^{-1}P^H(\mathbf{x}^n)\mathbf{\Phi_s}^n \quad = \quad P^H(\mathbf{x}^n)(P(\mathbf{x}^n)P^H(\mathbf{x}^n) + \lambda I)^{-1}\mathbf{\Phi_s}^n. \tag{18}$$

Usually, $N_s \times N_r$ is smaller than $N \times N$ in the simulation. In this case, the dimension of $(P(\mathbf{x}^n)P^H(\mathbf{x}^n) + \lambda I)$ is $(N_r \times N_s) \times (N_r \times N_s)$, while $(P^H(\mathbf{x}^n)P(\mathbf{x}^n) + \lambda I)$ is $N \times N$, so the computation time in inverting the matrices can be reduced using the right hand side expression in (18). As a matrix inverse routine, we used the complex Hermitian matrix inversion routine in the International Mathematical and Statistical Libraries (IMSL) .

5. NUMERICAL RESULTS

This section is devoted to several illustrative numerical results. Figure 1 shows the data acquisition geometry for all simulations. The optical modulation frequency used is 200MHz ($= \omega/2\pi$ in (1)). The unknown inhomogeneities are embedded in a homogeneous background medium with $\mu_a^b = 0.02cm^{-1}$ and $\mu_s'^b = 10.0cm^{-1}$, which is representative of tissue. A total of 12 sources and 12 detectors are located uniformly over the boundary of the $8 \times 8cm^2$ domain.

A multi-grid finite difference solver (MUDPACK[13]) is used for the forward solution of (1). The domain in Fig. 1 is discretized into 33×33 grid points and a zero fluence boundary condition is employed on the external boundary. The measured data are obtained at one grid point ($\frac{8}{32}cm$) inside this boundary. The finite difference solver provides a solution for the Green's function for the equivalent source at each point in the domain and also the incident fluence with an inhomogeneous k_b, thereby allowing $P(\mathbf{x}^n)$ and $\mathbf{\Phi_s}^n$ in (17) to be obtained. The initial guess for all reconstructions is the homogeneous background. If another reasonable starting point is used, similar results can be obtained for the image reconstruction. The goal is to simultaneously reconstruct the absorption and scattering coefficients at each of the grid points, which provides the desired image. Equation (17) is used in the reconstruction tests for different weighting matrices. The elapsed time for one iteration including both forward solution and inversion is about 1 minute on a Sun Sparc5 workstation.

5.1. Single Inhomogeneity

A change in only the absorption from the background is simulated in this case. Consider the geometry of Fig. 1 with a centrally located circular absorber of radius $0.7cm$ and $\mu_a = 0.2cm^{-1}$, which is 10 times larger than the background. We consider now three cases of (17): firstly, $W^n = I(q = 0)$ with $\beta^n = 1$ in (17), which is actually the DBIM with fixed λ, secondly, $W^n = I$ $(q = 0)$ with $\beta^n = 10$ in (17) and, finally, W^n given by (13) with $q = 1$ and again $\beta^n = 10$. In all the simulations, the parameters λ and β^n were fixed for all iterations to allow comparison of the different algorithms. The values of λ and β^n used were obtained empirically to ensure convergence.

The reconstructed μ_a and μ'_s for the three cases, the DBIM and the WDBIM with $\beta^n = 10$ and $q = 0, 1$, after 30 iterations, are shown in Fig. 2. The true images for μ_a and μ'_s are given in Fig. 2(a) and (b), where only μ_a varies. Figure 2 (c) and (d) show the quite poor reconstructed μ_a and μ'_s obtained using the DBIM with fixed λ. If the smaller values of λ are used, the algorithm diverges after a few iterations. The results for $q = 0$ in Fig. 2 (e) and (f) show improved results for μ_a, but not a quantitatively good reconstruction. The results for the case with $q = 1$ in Fig. 2 (g) and (h) show an accurate result for the peak in μ_a, a result which is clearly superior to the others. Also, note that the high frequency noise in μ'_s of Fig. 2(h) is somewhat smaller than that of Fig. 2(f). The reconstruction by the DBIM can be improved by using more iterations, but more than 200 iterations are required to obtain comparable quality to that obtained by the WDBIM.

5.2. Multiple Inhomogeneities

We now consider the case of multiple inhomogeneities. The inhomogeneities are again embedded in a homogeneous background medium with $\mu_a^b = 0.02cm^{-1}$ and $\mu_s'^b = 10.0cm^{-1}$. The inhomogeneous regions each have a radius of $0.7cm$ and their μ_a values are 10 times larger than those of the background, and μ_s is constant. The inhomogeneities are located at $(2.5cm, 2.5cm)$ and $(5.5cm, 5.5cm)$. The actual profiles for μ_a and μ'_s are shown in Fig. 3(a)-(b).

Figure 3(c)-(h) shows the reconstructed absorption and scattering distributions after 80 iterations using several algorithm parameters. Figure 3(c)-(d) show the reconstructed μ_a and μ'_s, respectively, with $W_n = I$ $(q = 0)$, $\beta = 1$, and fixed $\lambda = 0.1$ in (17). Smaller values of λ caused the algorithm to diverge. This corresponds to the DBIM with a fixed regularization parameter. Even though the inhomogeneities are observable, the quantitative values of the reconstruction are not good. Figure 3(e)-(f) are reconstructions using $W_n = I$ $(q = 0)$, $\beta = 10$ and $\lambda = 10^{-3}$. Quantitatively, the reconstruction results are improved over those of the DBIM with a fixed regularization parameter, but there is some high frequency noise in the reconstructed μ_a and μ'_s. If $q = 1$ and $\beta = 10$ is used for the weighting matrix computation, there is a noticeable improvement over the other reconstructions as can be seen in Fig. 3(g)-(h). In the reconstructed μ_a, the high frequency noise is significantly reduced using the WDBIM. However, a considerable amount of noise is seen in the reconstructed μ'_s, although in Fig. 3(h), the magnitude is significantly reduced compared to that of Fig. 3(f). The noise may be caused by the significant imbalance between the μ_a and μ'_s variations in the actual profile. As in the case of a single inhomogeneity, more than 200 iterations of the DBIM are required to obtain reconstructions comparable to those using the WDBIM shown in Fig. 3(g)-(h).

5.3. Simulated Phantom

We now consider the case of a simulated phantom whose true image is shown in Fig. 4(a)-(b). Even though it has a complicated structure, our algorithm can reconstruct its overall structure with only 12 sources and 12 detectors. The quantitative values of the peaks for each inhomogeneity are restored quite accurately, as seen in Fig. 4(c)-(d). The blurring is due partly to the limited number of sources and detectors.

6. CONCLUSION

A weighted distorted Born iterative method has been developed which provides faster convergence for reconstruction of optical diffusion images relative to those achieved using the standard distorted Born method with a fixed regularization parameter. This approach allows variable weighting and step size during the iterative solution. The special case of uniform weighting and unit step size gives the standard distorted Born iterative method. Numerical experiments with single and multiple inhomogeneities with high contrast show that the weighted approach produces more accurate images than does the distorted Born iterative method, for a small number of iterations.

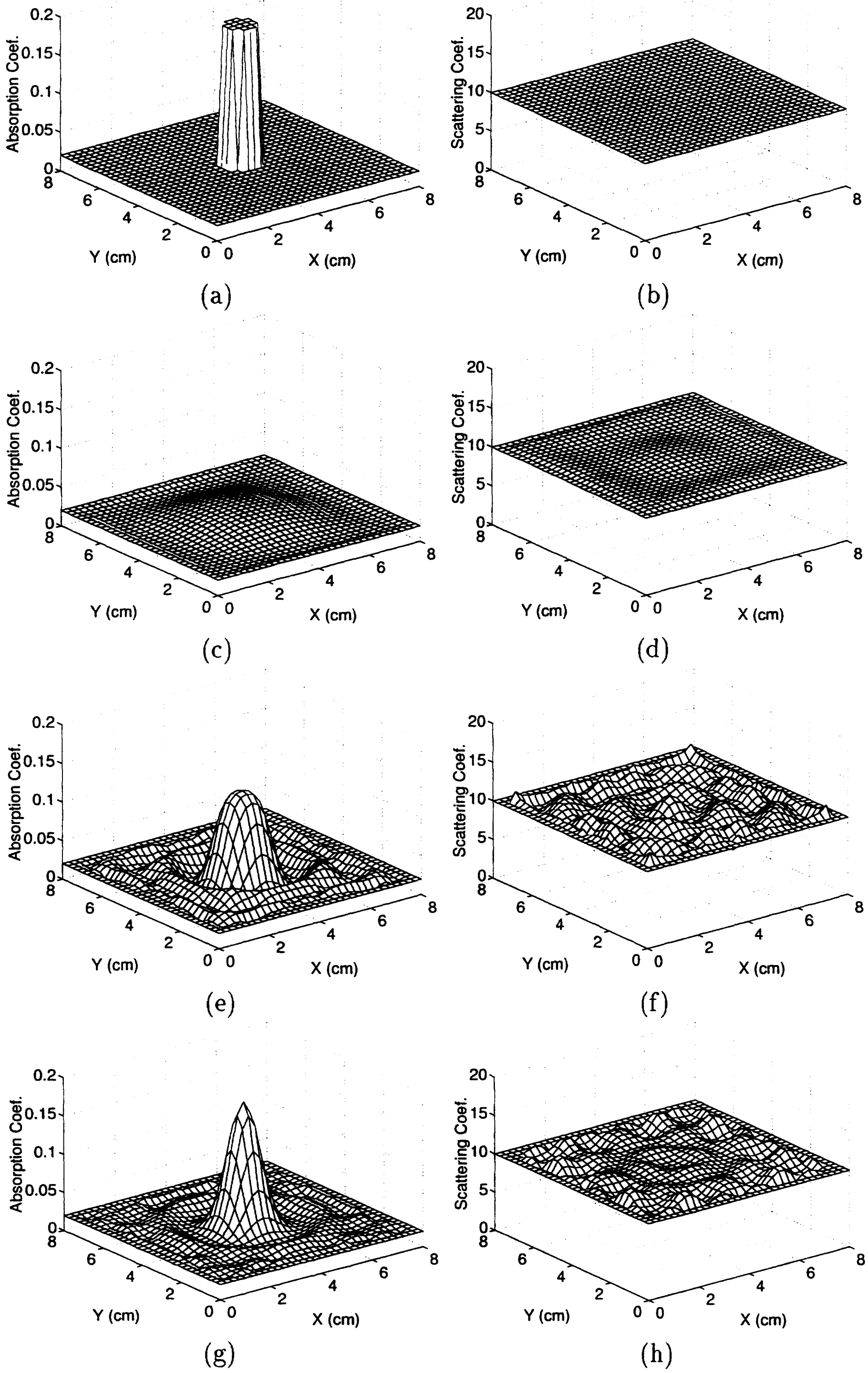

Figure 2. Comparison of the reconstructed μ_a and μ_s' of a centrally-located circular absorber with $\mu_a = 0.2cm^{-1}$, after 30 iterations by several algorithms: (a)-(b) actual profile of μ_a, μ_s'; (c)-(d) results by the DBIM with fixed $\lambda = 0.1$; (e)-(f) uniform weighting ($q = 0$), $\beta^n = 10$, $\lambda = 1.0 \times 10^{-8}$; (g)-(h) $q = 1$, $\beta^n = 10$, $\lambda = 1.0 \times 10^{-8}$.

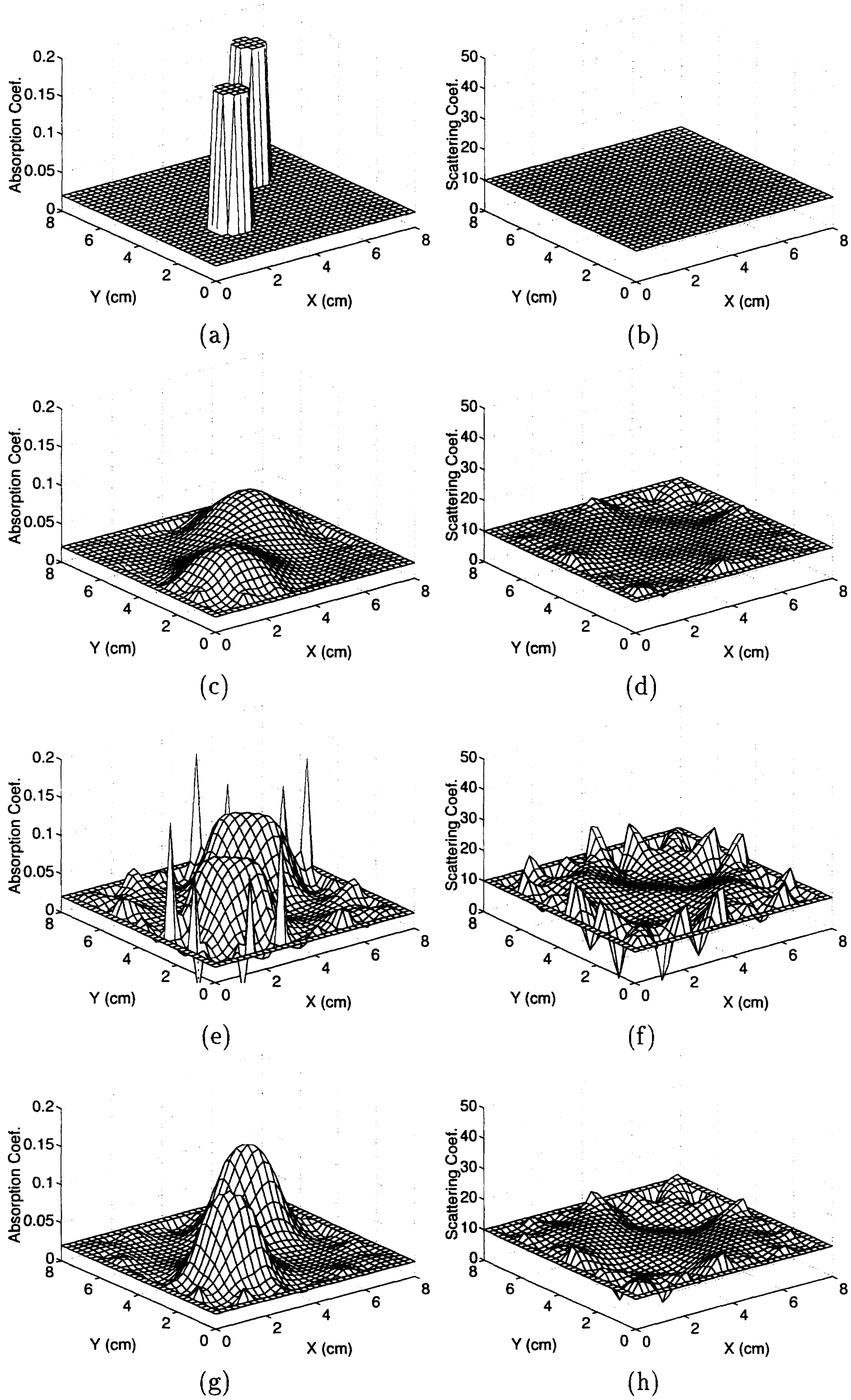

Figure 3. Comparison of the reconstructed μ_a and μ_s' of two circular absorbers with $\mu_a = 0.2cm^{-1}$, which are located at $(2.5cm, 2.5cm)$ and $(5.5cm, 5.5cm)$, after 80 iterations by several algorithms: (a)-(b) actual profile of μ_a, μ_s'; (c)-(d) results by the DBIM with fixed $\lambda = 0.1$; (e)-(f) uniform weighting $(q = 0)$, $\beta^n = 10$, $\lambda = 1.0 \times 10^{-3}$; (g)-(h) $q = 1$, $\beta^n = 10$, $\lambda = 1.0 \times 10^{-3}$.

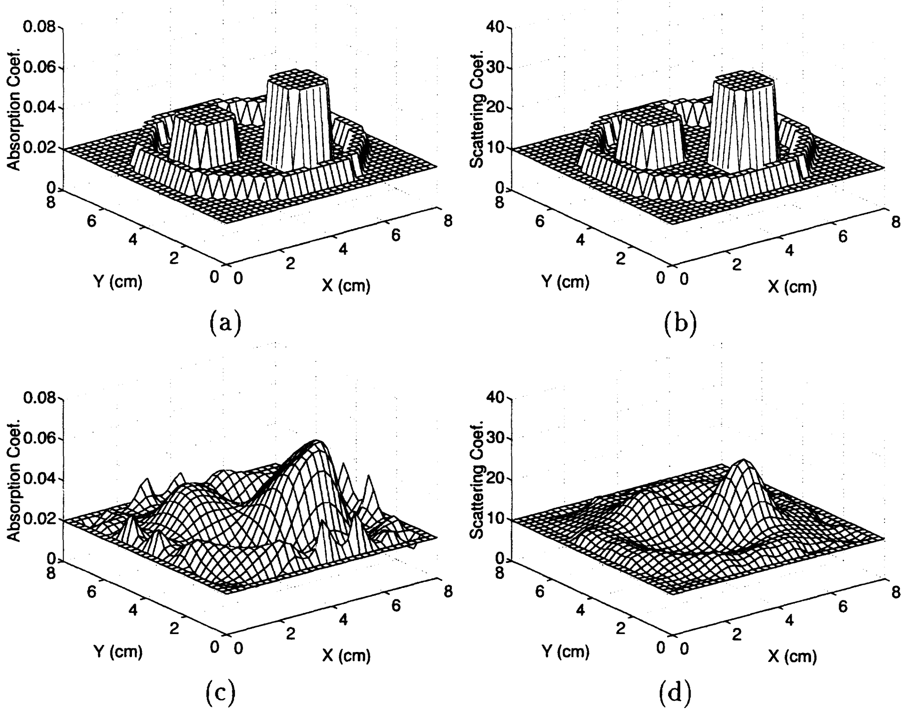

Figure 4. Reconstructed μ_a and μ_s for a phantom model with three inhomogeneities after 200 iterations. (a)-(b) Actual μ_a and μ_s; (c)-(d) WDBIM reconstruction with $q = 1, \beta = 10, \lambda = 1.0 \times 10^{-4}$.

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